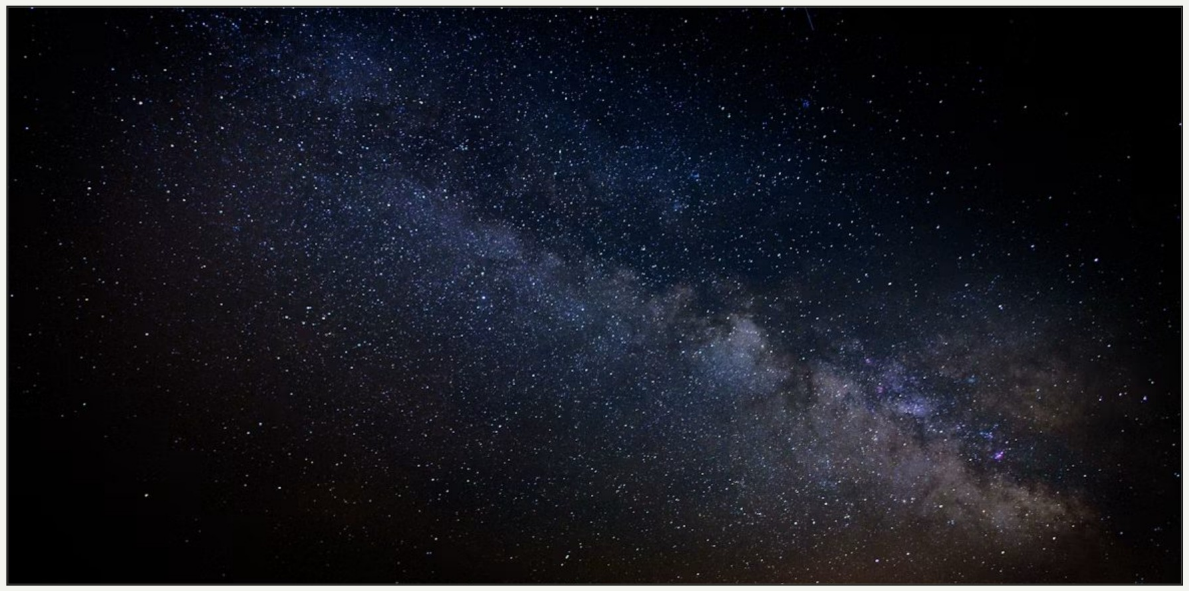
Art direction reference

Images supplied for mood. Reference only: the game ships original low-poly assets, and none of these nine has a verified licence.

LANDING BACKDROP · OPTIONS FOR SCREEN 1



A · Moon with Earth as a crescent. Used on screen 1A.



B · Milky Way night sky. Used on screen 1B.

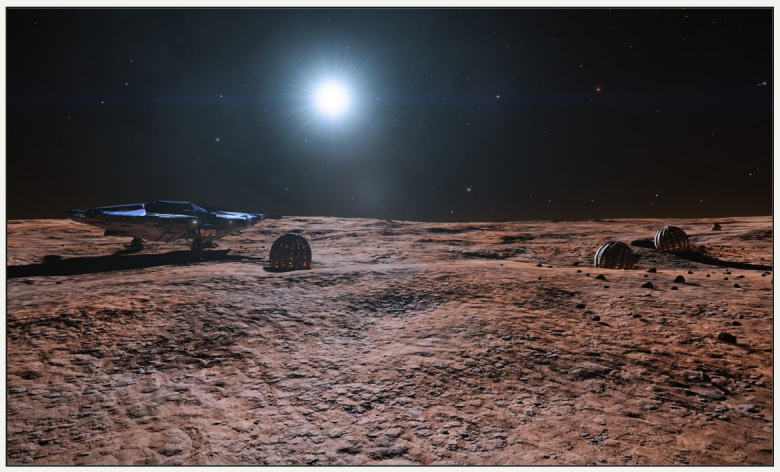
TERRAIN LOOK



1 · Grey dust, low hills, black sky with Earth.

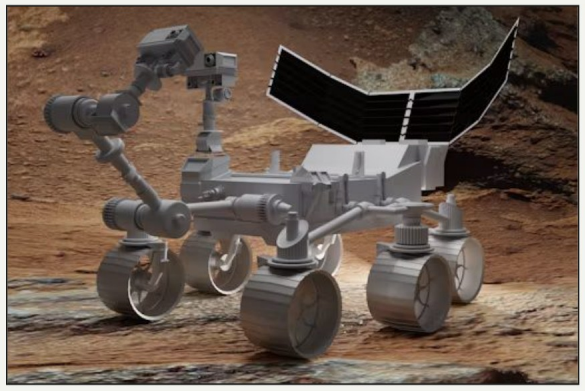


2 · Red rock and cracked ground, hazy sky, jagged peaks.



3 · Rocky red-brown ground, bright star, scattered stones.

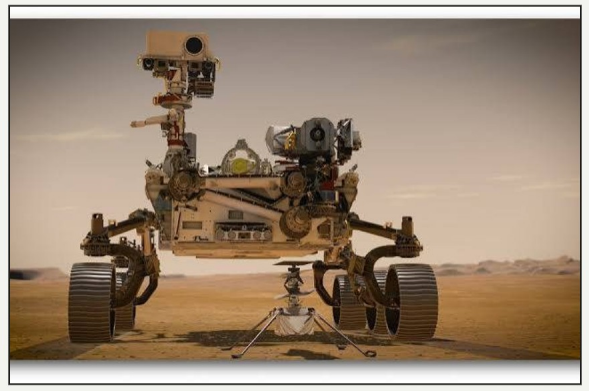
ROVER LOOK



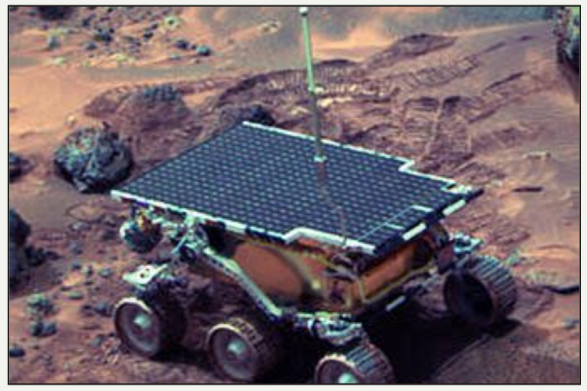
1 · Six wheels, camera mast, solar wing.



2 · Orange body, big treaded wheels, light bar.



3 · Front view, mesh wheels, exposed chassis.



4 · Small body, flat solar deck, six wheels.

What the look must do for the game

Show the three slots on the model: chassis, wheels, motor.

Make the three wheel types look different (smooth, all-terrain, spiked), because wheels are a physics part.

Make the surfaces readable from the driving view: dark paved strip, pale sand, pale blue ice.

Proposed terrain theme

A Mars-analog proving ground on Earth: red ground and sky, with a paved test track. Gravity stays 9.8 m/s² (SRS 2.2). The Moon and alien looks fit later levels, where a planet's gravity would be a modifier (SRS Section 6).

Build a rover. Predict. Drive. Learn.

Grade 9 Kinematics and Dynamics, one rover at a time.

Email

name@school.example

Password

Password

Sign in | Create account

Forgot password?

or

Play as guest

Guests save progress on this device only. An account saves progress to the cloud, so it follows the student to another device.

FR-25 · FR-41



Backdrop option A: Moon and Earth (reference image)

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Grade 9 Kinematics and Dynamics, one rover at a time.

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name@school.example

Password

Password

Sign in

Create account

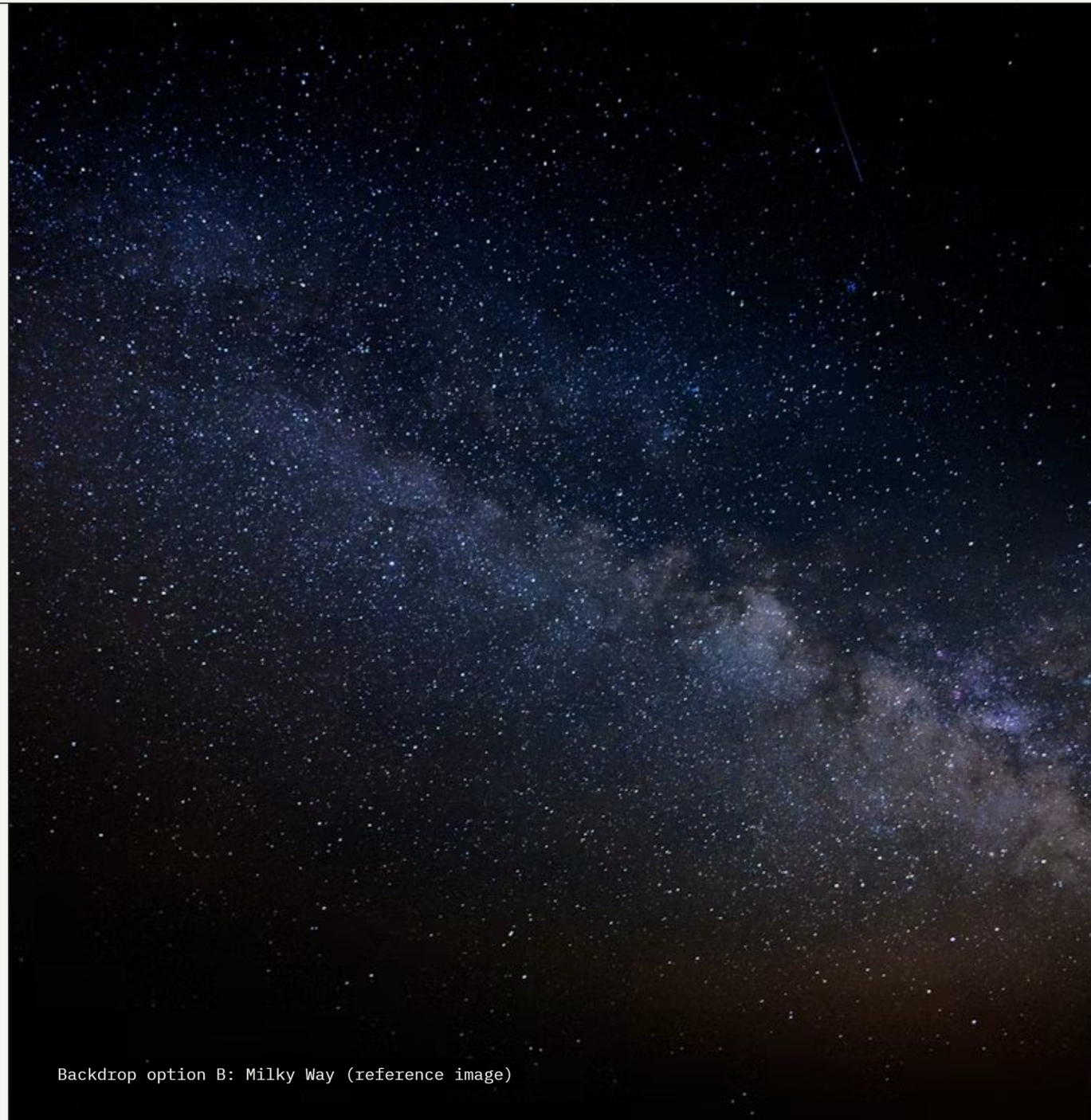
Forgot password?

or

Play as guest

Guests save progress on this device only. An account saves progress to the cloud, so it follows the student to another device.

FR-25 · FR-41



Choose a mission

One terrain, three missions. Finish one to open the next.

FR-12 · FR-36

MISSION 1

Motion Trial



Concept: distance, time, speed, acceleration

From rest, reach the target within 9 s: take the longer route on fast ground, or a shortcut across slow ground.

Available

Start

MISSION 2 · LOCKED

Surface Test



Concept: friction: grip and rolling friction

Cross the ice strip in time, then coast far enough into the sand.

Needs Mission 1 completed and 20 Knowledge.

Unlock · 20 Knowledge

MISSION 3 · LOCKED

Stopping Challenge



Concept: force, friction and speed combined

Towed at a set speed on asphalt, brake after the line and stop with the whole rover inside the target zone.

Needs Mission 2 completed and 20 Knowledge.

Unlock · 20 Knowledge

Knowledge is earned by clearing questions, and spent on missions and rover parts.

Concepts cleared

Account

Your rover



Chassis: Standard
Wheels: All-terrain
Motor: Medium

Mission 1 routes: asphalt 100 m, sand shortcut 85 m
Time limit 9 s · starts from rest (0 m/s)

FR-4

Choose one part per slot

FR-1 · FR-36 · FR-37

CHASSIS

Light
120 kg
Unlock · 15

Standard
160 kg
Selected · starting part

Heavy
220 kg
Unlock · 15

WHEELS

Smooth
16 kg
grip ×0.8 · rolling ×0.6
Unlock · 15

All-terrain
24 kg
grip ×1 · rolling ×1
Selected · starting part

Spiked
32 kg
grip ×1.4 · rolling ×1.6
Unlock · 15

MOTOR

Small
15 kg · 700 N
top speed 18 m/s
Unlock · 15

Medium
25 kg · 1000 N
top speed 16 m/s
Selected · starting part

Large
40 kg · 1400 N
top speed 14 m/s
Unlock · 15

Answer an unlock question · +5 Knowledge

Opens after Mission 1's required questions are answered.

FR-34

Part effects

FR-2 · FR-3

	Asphalt	Sand
Mass	209 kg	209 kg
Weight	2048 N	2048 N
Rolling friction	41 N	246 N
Grip limit	1639 N	1024 N
Driving force	1000 N	1000 N
Top speed	16 m/s	16 m/s
Max acceleration	4.59 m/s²	3.61 m/s²
Max braking	7.37 m/s²	6.08 m/s²

Values update when a part changes. They come from the same physics model that drives the run. Surfaces shown are the ones in this mission.

Continue to questions

3 required questions to answer before the run

REQUIRED QUESTION 1 OF 3 · PREDICTION

FR-17 · FR-18 · FR-19

Rover A and Rover B have the same driving force. Rover A has twice the mass of Rover B. Ignoring friction, which statement is correct?

A Rover A accelerates twice as fast as Rover B.

B Rover B accelerates twice as fast as Rover A.

C They accelerate equally, because the force is equal.

D Rover B accelerates four times as fast as Rover A.

Clear it to earn 10 Knowledge. Sample question for layout; the real questions are still to be written.

Reset wireframe

Wrong answer: the explanation shows, Close stays locked for 2 s (FR-21), and the question returns later as a variant.

Your build

rover preview (placeholder)

Chassis: Standard
Wheels: All-terrain
Motor: Medium
Mass: 209 kg

Back to build

Predict your run

FR-26 · FR-32

From rest, reach the target within 9 s. Choose your route while you drive: the longer way on fast ground, or the shortcut across slow ground.

TWO ROUTES, DRAWN TO SCALE

START Longer route · 100 m · fast ground TARGET

START Shortcut · 85 m · slow ground TARGET

Each route starts with two pickups side by side, Motor boost and Lighter load. Only one can be active.

How long will your rover take to reach the target?

- ◯ Under 8 s
- ◯ 8 s to 9 s
- ◯ More than 9 s, so it misses the limit

This prediction earns no Knowledge. The results screen shows it next to what happened.

Start run

TIME
3.2 s
limit 9 s

Motor boost active
Driving force 1000 N → 1200 N
FR-38 · FR-40

SPEED
9.4 m/s

FR-27

sample values for layout

3D view: terrain per the art direction board

rover
seen from behind

Target in 57 m
markers every 5 m

Throttle 60%
Brake 0%

Levels rise while a key is held and fall when released. Steer left and right. Key choice is a proposal.

FR-6 · FR-7

Wheels slipping: off FR-9

TIME

2.1 s

paused

PICKUP · MOTOR BOOST · OPTIONAL Timer paused

FR-33 · FR-38 · FR-40

The driving force on a rover goes up by 20% and its mass stays the same. Ignoring friction, what happens to its acceleration?

A It goes up by 20%.

B It goes down by 20%.

C It stays the same, because the mass is unchanged.

D It doubles.

If correct: Motor boost for this attempt. Driving force 1000 N → 1200 N. +5 Knowledge the first time.

If wrong: no penalty. The explanation shows, then the run continues.

Sample question for layout; the real questions are still to be written.

Mission complete: target reached in 8.50 s

Limit 9 s. Progress saved: a completed mission is the safe point.

FR-14 · FR-15 · FR-16 · FR-24

Measured results

Distance	**100 m**
Time	**8.50 s**
Average speed	**11.76 m/s**
Top speed reached	*[from the run] m/s*
Average acceleration	*[from the run] m/s²*

Your prediction and what happened

Predicted time	**8 s to 9 s**
Actual time	**8.50 s**

Sample run for layout: starting build, longer route.

Why it happened

You covered 100 m in 8.50 s, inside the 9 s limit. Average speed is the distance divided by the time.

```
average speed = distance ÷ time
= 100 m ÷ 8.50 s
= 11.76 m/s
```

The shortcut is only 85 m, but it crosses slow ground, where the rover's speed rises more slowly. A shorter route is not always the quicker one.

Acceleration is net force divided by mass (a = F ÷ m). With 209 kg, your rover's speed rose by up to 4.59 m/s every second on fast ground. A heavier rover, or a smaller motor, accelerates less and takes longer.

Pre-written explanation filled in with this run's values. Mission 1 uses distance, time, speed, acceleration and F = ma; friction waits for Mission 2.

Unlock Mission 2 · 20 Knowledge

30 Knowledge now, 10 left after.

Retry mission **Change build** **Concepts cleared**

By mission

Mission 1 · 2 cleared

Mission 2 · locked

Mission 3 · locked

Back to missions

Mission 1 · Motion Trial

FR-20 · FR-21 · FR-22

Required question 1 · force, mass and acceleration Cleared · +10

Acceleration is net force divided by mass (a = F / m). With the same force, half the mass gives twice the acceleration.

Required question 2 · [title to be written] Cleared · +10

[Explanation to be written with the question.]

Required question 3 · [title to be written] To review

Answered wrongly. It comes back as a variant: the same concept with a different scenario or numbers.

Only cleared questions list their explanations. Questions not yet answered stay hidden. Sample state for layout.

UNLOCK QUESTION · MISSION 1 CONCEPT · OPTIONAL

FR-34 · FR-35

A 300 kg rover has a net force of 1200 N. A new motor raises the net force to 1800 N. The mass does not change. By how much does the acceleration increase?

A By 2 m/s²

B By 6 m/s²

C By 1.5 m/s²

D It stays the same, because the mass is unchanged.

Worth 5 Knowledge the first time it is cleared. Wrong answers cost nothing; the question comes back as a variant.

Back to build

Sample question for layout; the real questions are still to be written.